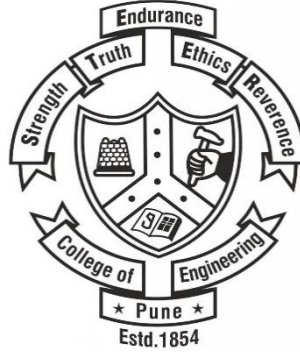




College of Engineering Pune

(An Autonomous Institute of the Govt. of Maharashtra)

YEAR : SECOND YEAR BACHELOR OF PLANNING
SEMISTER: IV

SUBJECT: Geo-informatics-I

Prepared by,

Jaysing Jadhav

***Adjunct Faculty, Planning section,
Dept. of Civil Engineering CoEP***

Course Outcomes:

CO1: Understand basics of remote sensing in practical sense.

CO2: Visual interpretations of aerial imagery.

CO3: Analysis base for big data acquired from remote sensing and planning sources.

CO4: Computational interpretation of aerial imagery.

CO5: Data handling systems information in India.

(SBC1) () Geo-informatics - I	
Teaching Scheme	Examination Schemes
Lectures: 4 Hrs./week	T1 and T2 - 20 Marks each
	End Sem- 60 Marks

UNIT 1: REMOTE SENSING Limitations of Traditional Surveys for Planning; Remote Sensing - Definition, Aerial and Satellite Remote Sensing, Aerial Remote Sensing	(2 Hrs)
UNIT 2: PHOTO INTERPRETATION Aerial Photo-Interpretation, Qualitative and Quantitative Elements of Photo-Interpretation; Satellite Remote sensing, Geo-Stationary and Sun-Synchronous Satellites, Principles of Electro-Magnetic Radiations, Resolutions; Introduction to Digital Image Processing; Salient Features of Popular Remote Sensing Satellites; Applications in Planning; Laboratory Exercises	(4 Hrs)
UNIT 3: PLANNING INFORMATION SYSTEMS Systems Approach to Planning as basis for Planning Information Systems; Systems, Hierarchy, Types; Data and Information, Value of Information, Information Flows, Loops; Information Security and Sharing; Information Systems, Types, Limitations	(4 Hrs)
UNIT 4: IMAGE ANALYSIS AND SOFTWARE APPLICATIONS ERDAS Imagine software and its applications in planning, Image Acquisition- BHUVAN, USGS and concerned software applications (QGIS). Project based informatics typology and analysis methods for; Asset management (Land use Land cover), Property appraisal (Change detection), Vegetation indexing (NDVI feature extraction, mangroves wetland-loss and encroachment), Disaster Mapping (Mapping flood extent areas and disaster impact zones), Mining operations (Pinpoint the location of potential mineral outcroppings through remote sensing. Monitor active or inactive mines and understand broad environmental impacts	(12 Hrs)
UNIT 5: PLANNING INFORMATION SYSTEMS IN INDIA Planning Information Systems -NNRMS, NUIS, National Urban Observatory, Municipal Information Systems, Land Information Systems, Cadastre Systems; Applications and Limitations; Tools for Spatial Data Handling, Introduction to GIS. GIS based master planning	(3 Hrs)

Reference Books:

•Bhatta, B. (3 Nov 2011). Remote Sensing and GIS (ISBN 019569239X, 9780195692396 ed.). University of Minnesota: Oxford University Press, 2008.

•Reddy, M. A. (2001). Textbook of Remote Sensing and Geographical Information Systems (ISBN 8178000210, 9788178000213 ed.). BSP BS Publication.

•HarsanKarimi (Ed.) (2009) Handbook of Research on Urban Informatics: The Practice and Promise of the Real-Time City. Information Science Reference, IGI Global, Hershey, Pa.

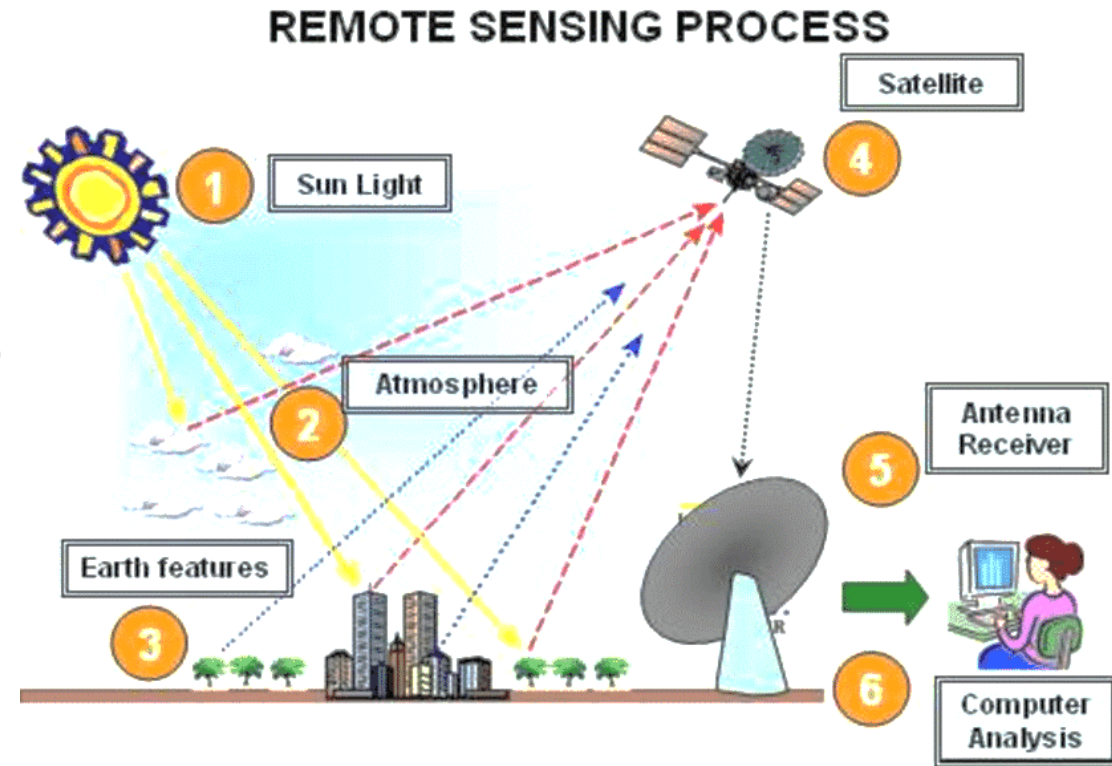
•M. S. Nathawat, A. C. Pandey (Eds.) (2008) Geoinformatics for Decentralized Planning and Governance (ISBN: 813160117X), Rawat Publications

REMOTE SENSING

- ❑ Remote sensing is a method of obtaining information about the properties of an object without coming into physical contact with it.
- ❑ Remote sensing is the process of detecting and monitoring the physical characteristics of an area by measuring its reflected and emitted radiation at a distance (typically from satellite or aircraft). Special cameras collect remotely sensed images, which help researchers "sense" things about the Earth.

Some examples are:

- ❑ Cameras on satellites and airplanes take images of large areas on the Earth's surface, allowing us to see much more than we can see when standing on the ground.
- ❑ Sonar systems on ships can be used to create images of the ocean floor without needing to travel to the bottom of the ocean.
- ❑ Cameras on satellites can be used to make images of temperature changes in the oceans.



Some specific uses of remotely sensed images of the Earth include:

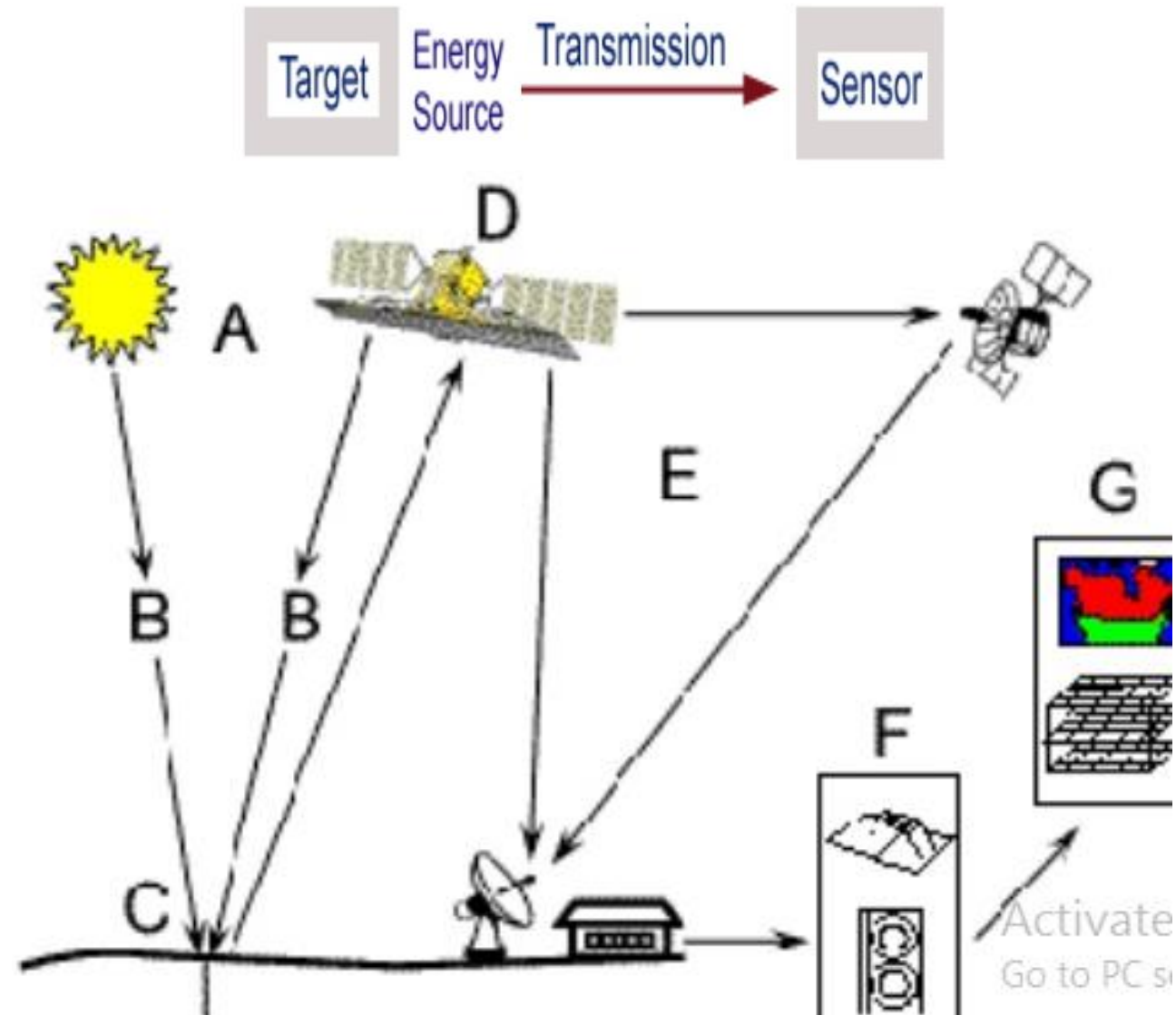
- ❑ Large forest fires can be mapped from space, allowing rangers to see a much larger area than from the ground.
- ❑ Tracking clouds to help predict the weather or watching erupting volcanoes, and help watching for dust storms.
- ❑ Tracking the growth of a city and changes in farmland or forests over several years or decades.
- ❑ Discovery and mapping of the rugged topography of the ocean floor (e.g., huge mountain ranges, deep canyons, and the “magnetic striping” on the ocean floor).



Remote Sensing System capture radiation in different wavelength reflected/ emitted by the earth's surface features and recorded it either directly on the film as in case of aerial photography or in digital medium used for generating the images RS provides valuable data over vast area in a short time about resources, meteorology and environment leading to better resource management and accelerating national development

Elements in Remote sensing

1. Energy source or Illumination (A)
2. Radiation the Atmosphere (B)
3. Interaction with the object (C)
4. Recording of energy by the sensor (D)
5. Transmission, reception & processing (E)
6. Interpretation & analysis (F)
7. Application (G)



ADVANTAGES OF REMOTE SENSING

- ❑ Provides a regional view (large areas).
- ❑ Provides repetitive looks at the same area.
- ❑ Remote sensors "see" over a broader portion of the spectrum than the human eye.
- ❑ Sensors can focus in on a very specific bandwidth in an image or a number of bandwidths simultaneously.
- ❑ Provides geo-referenced, digital, data.
- ❑ Some remote sensors operate in all seasons, at night, and in bad weather
- ❑ Overcomes inaccessibility
- ❑ Rapid extraction of GIS ready data

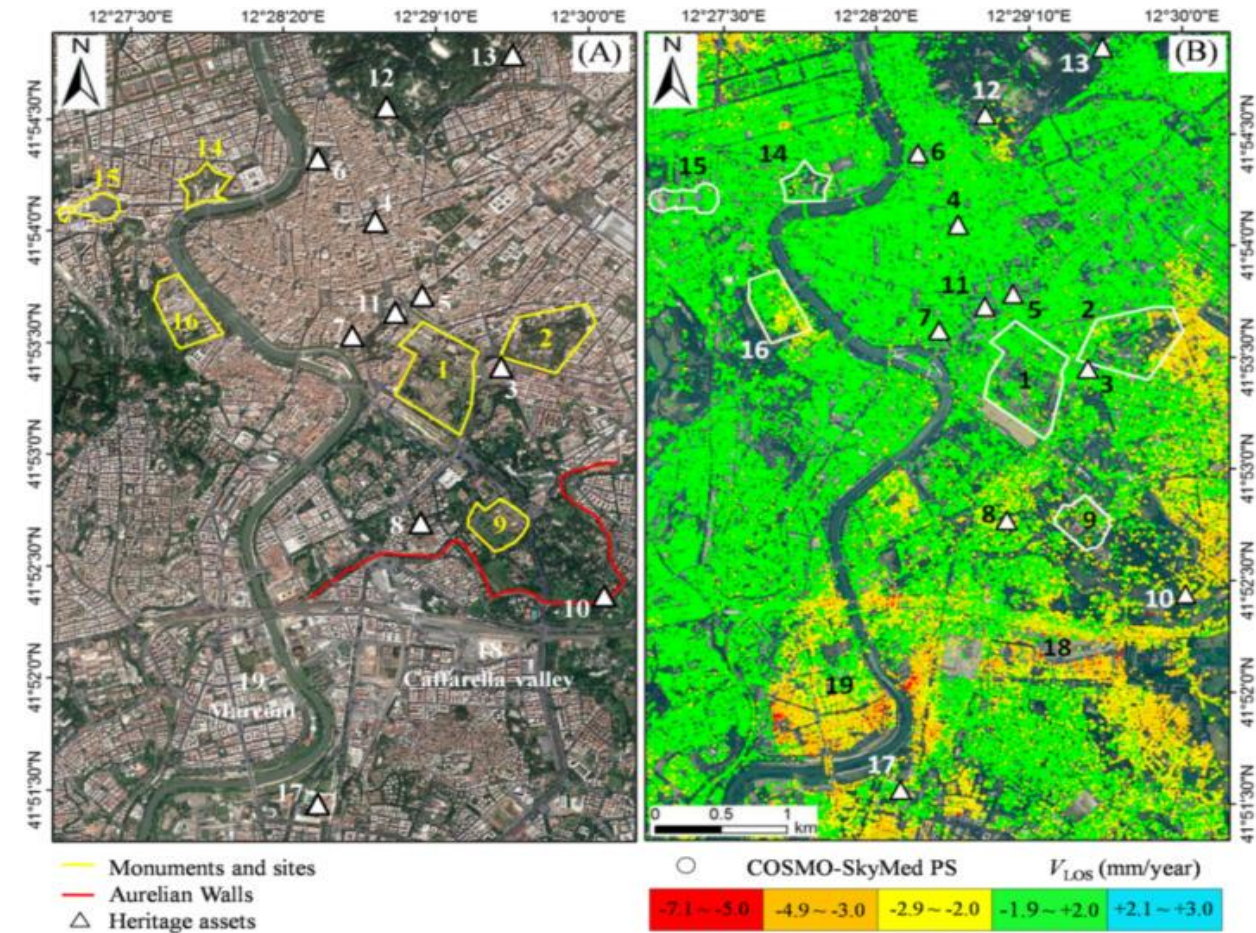
DISADVANTAGES OF REMOTE SENSING

- ❑ Needs ground verification
- ❑ Limitations on offering details
- ❑ Not the best tool for small areas
- ❑ Need expert system to extract data
- ❑ High-resolution data from remote sensing can be hard to store



Why Remote sensing over traditional surveying ?

- ❑ Consistent, routine, global measurements
- ❑ Overview of information on the hemispheric, regional , national & local scales- the big picture
- ❑ Provide information in areas where there are no ground based measurements
- ❑ Advance warning of impending environmental events and disasters
- ❑ Visual appeal; a picture is worth of thousand words
- ❑ Great speed of data collection
- ❑ Cheap and easily available
- ❑ Remote Sensing Doesn't Disturb People or the Environment



DATA MODELS: RASTER & VECTOR DATA MODEL

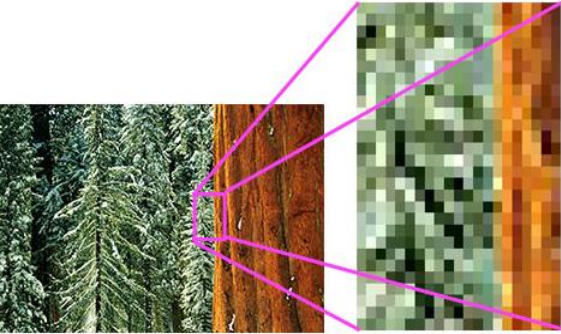
- ❑ Spatial data in GIS has two primary data formats: raster and vector.
- ❑ Raster uses a grid cell structure, whereas vector is more like a drawn map.
- ❑ Vector format has points, lines, polygons that appear normal, much like a map
- ❑ Raster format generalizes the scene into a grid of cells, each with a code to indicate the feature being depicted. The cell is the minimum mapping unit
- ❑ Raster has generalized reality: all of the features in the cell area are reduced to a single cell identity

RASTER DATA

- ❑ Raster is a method for the storage, processing and display of spatial data.
- ❑ Each area is divided into rows and columns, which form a regular grid structure.
- ❑ Each cell must be rectangular in shape, but not necessarily square.
- ❑ Each cell within this matrix contains location co-ordinates as well as an attribute value.
- ❑ The origin of rows and column is at the upper left corner of the grid.
- ❑ Rows function as the Y-coordinate and column as X-coordinate in a two dimensional system.
- ❑ A cell is defined by its location in terms of rows and columns.

157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
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199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
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190	214	173	66	103	143	96	50	2	109	249	215
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195	206	123	207	177	121	123	200	175	13	96	218



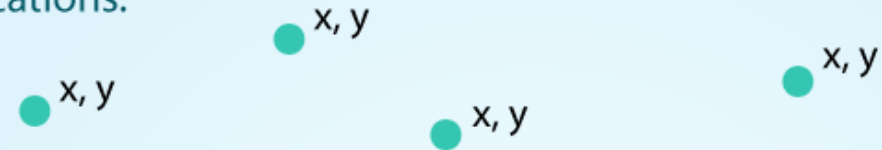
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183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

VECTOR DATA

- ❑ Vector data uses two dimensional Cartesian coordinates to store the shape of spatial entity.
- ❑ Vector based features are treated as discrete geometric objects over the space.
- ❑ In the vector data base point is the basic building block from which all the spatial entities are constructed.
- ❑ The vector spatial entity, the point is represented by a single x,y coordinate pair.
- ❑ Line and area entities are constructed by a series of points into chains

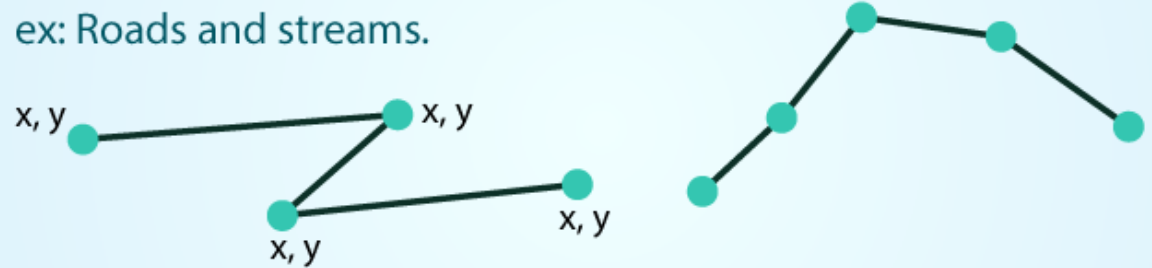
POINTS: Individual x, y locations.

ex: Center point of plot locations, tower locations, sampling locations.



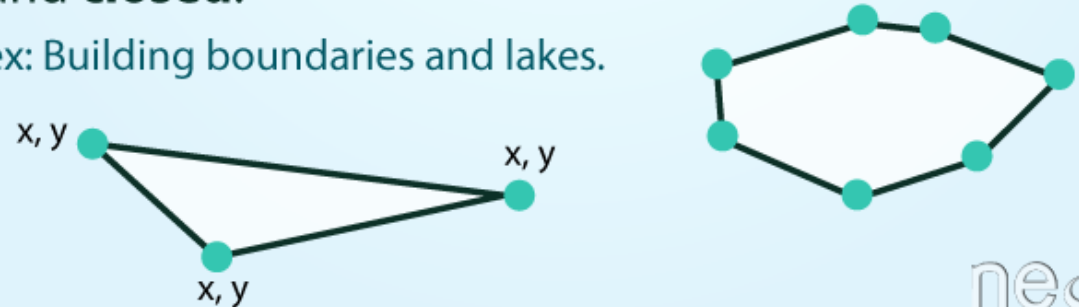
LINES: Composed of many (at least 2) vertices, or points, that are connected.

ex: Roads and streams.



POLYGONS: 3 or more vertices that are connected and **closed**.

ex: Building boundaries and lakes.

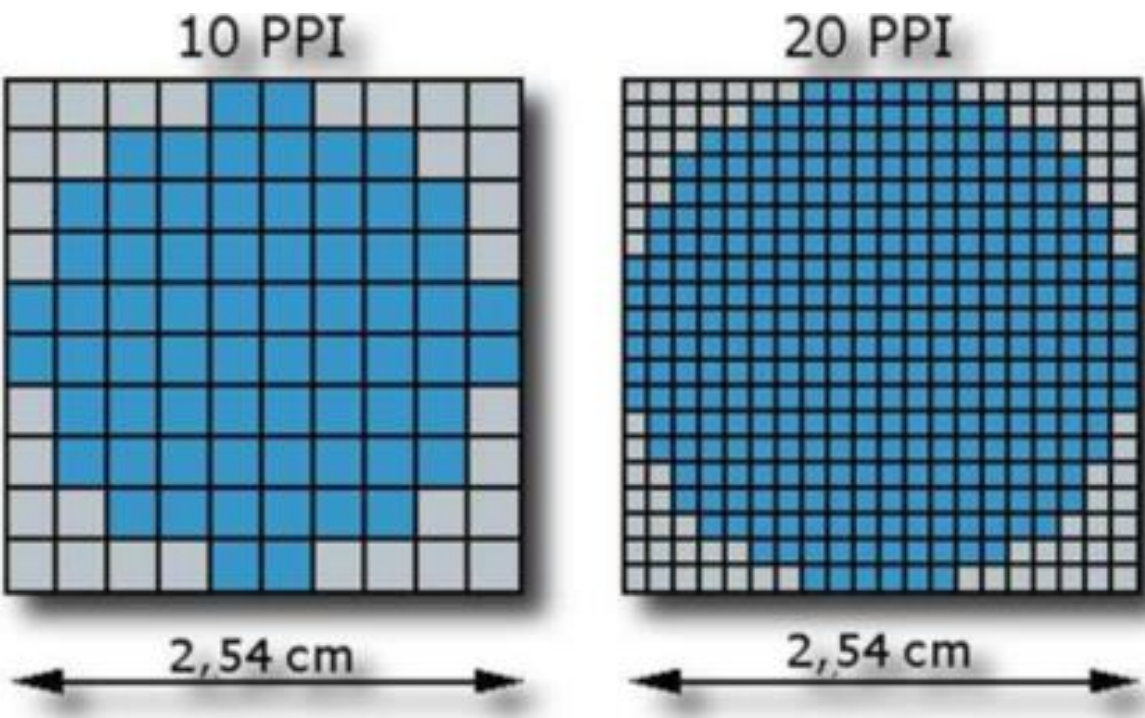


What is Pixel ?

A pixel is the smallest unit of a digital image or graphic that can be displayed and represented on a digital display device.

A pixel is the basic logical unit in digital graphics. Pixels are combined to form a complete image, video, text, or any visible thing on a computer display.

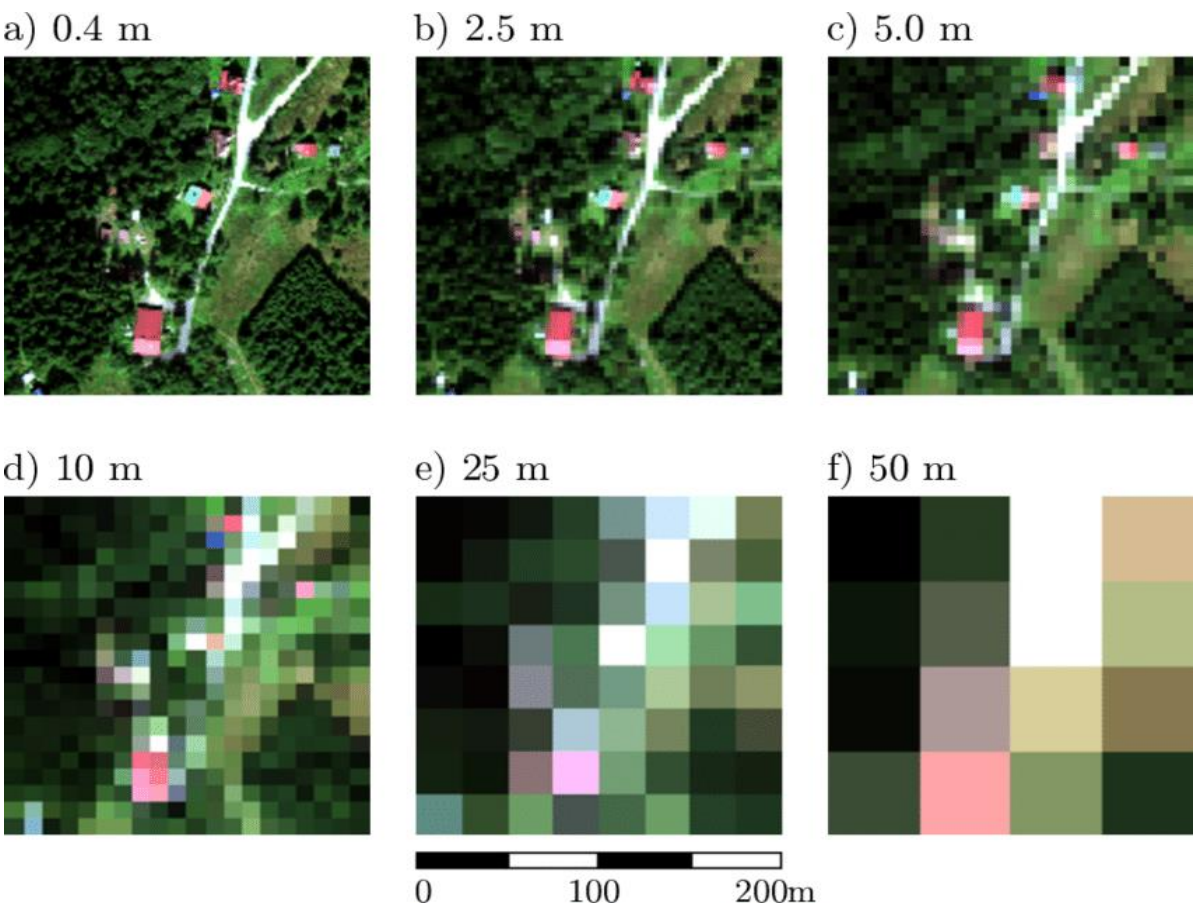
A pixel is also known as a picture element (pix = picture, el = element).



What is Image resolution?

Image resolution is typically described in PPI, which refers to how many pixels are displayed per inch of an image.

Higher resolutions mean that there more pixels per inch (PPI), resulting in more pixel information and creating a high-quality, crisp image



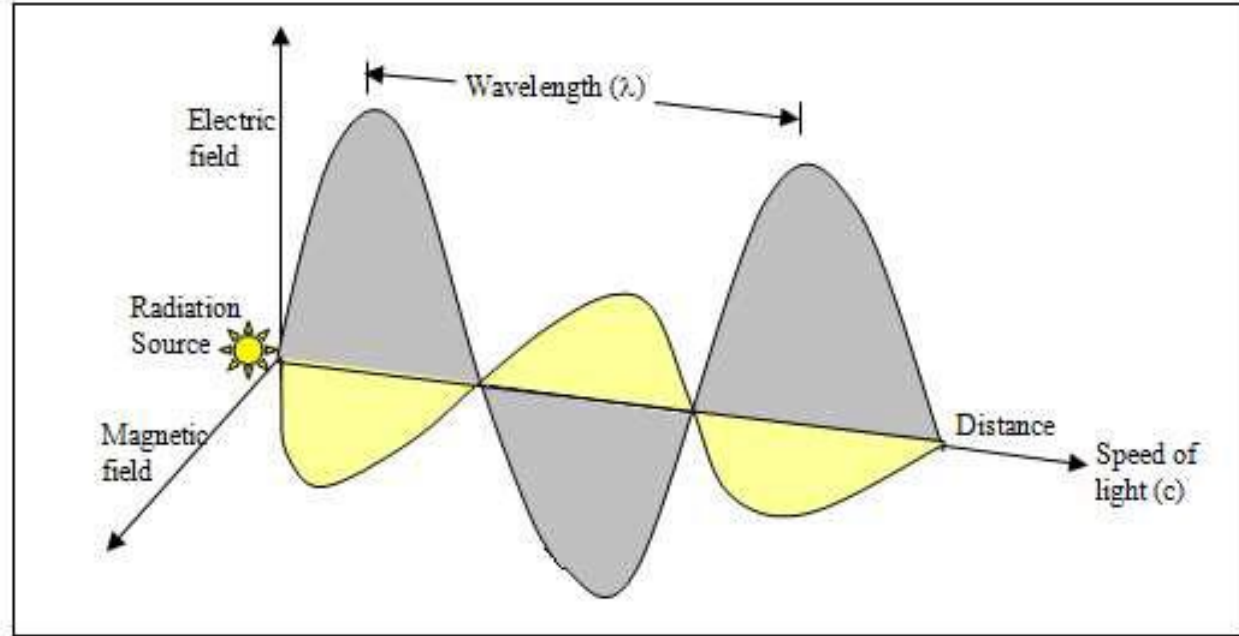
THE ELECTROMAGNETIC RADIATION

Electromagnetic radiation (EMR), also called as electromagnetic energy, and refers to all energy that moves with the velocity of light in the form of waves.

The source of EMR is the subatomic vibration of photons and is measured in terms of wavelength.

Sun is the main source of EMR that travels through space in the form of waves that are either reflected or absorbed by the objects, mainly on account of the size of wavelength which is described by the distance of successive wave peaks and is represented by a Greek letter lambda

Frequency refers to the number of cycles of a wave passing a fixed point per unit of time and is normally measured in hertz (Hz). Wavelength and frequency are inversely related to one another, in other words as one increases the other decreases. This relationship is expressed as:



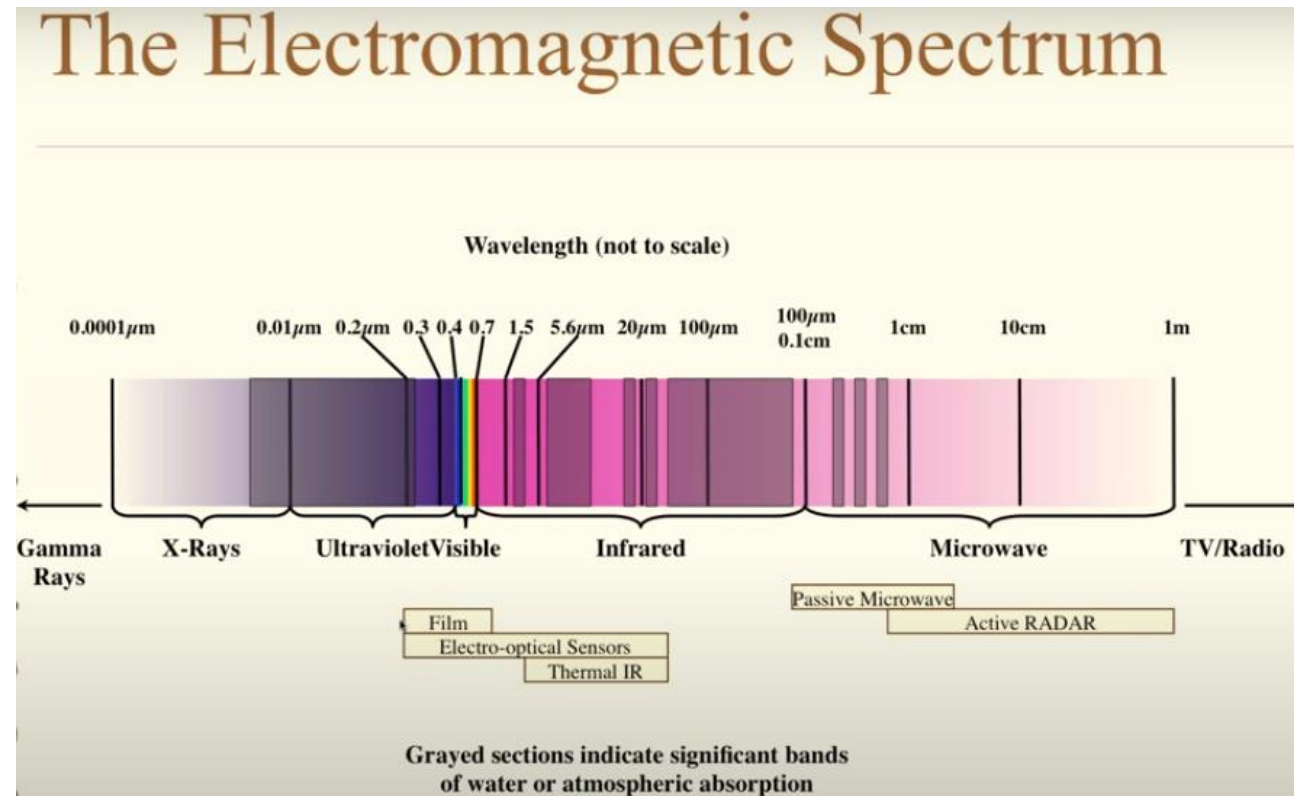
$$C = \lambda \nu$$

Where: c = speed of light (3×10^8 m/s), λ is the wavelength in (m, cm, μm or nm) and ν is the frequency in Hertz (cycles/second)

ELECTROMAGNETIC SPECTRUM

The electromagnetic spectrum covers the entire range of photon energies arranged in the increasing order of wavelengths on a logarithmic scale .

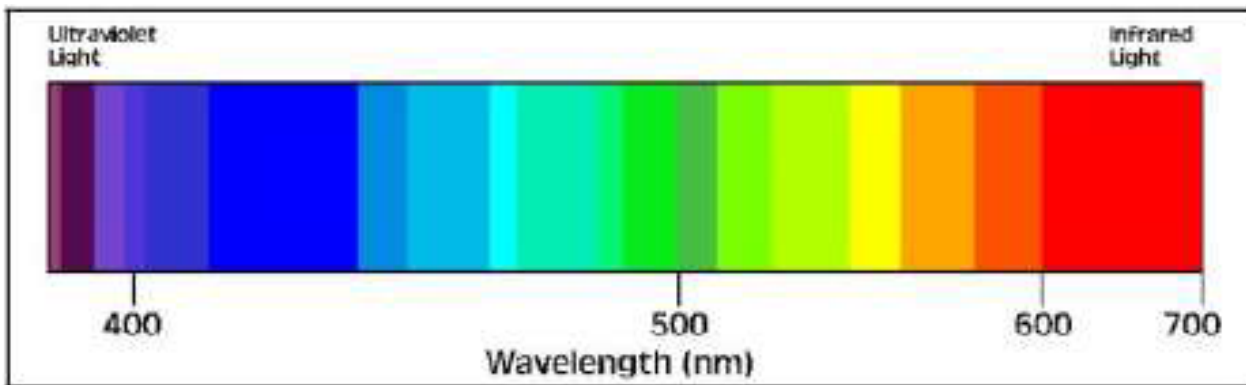
The electromagnetic spectrum ranges from the shorter wavelengths (including gamma and X rays) to the longer wavelengths (including microwaves and radio waves).



VISIBLE LIGHT:

This is the portion of EMR to which our eyes are sensitive and is perceived as colors.

The visible light covers a very small portion of the spectrum, ranging from approximately 0.4 to 0.7 μm (micrometer) and due to this small wavelength range, the visible portion of the spectrum is plotted on a linear scale so that individual colors can be discretely depicted. The visible light comprises of discrete colors with red at the long wavelength end and violet at the short wavelength.



The visible spectrum.

The visible colors and their corresponding wavelengths are listed below in micrometers (μm).

Violet: 0.4 to 0.446 μm

Blue: 0.446 to 0.500 μm

Green: 0.500 to 0.578 μm

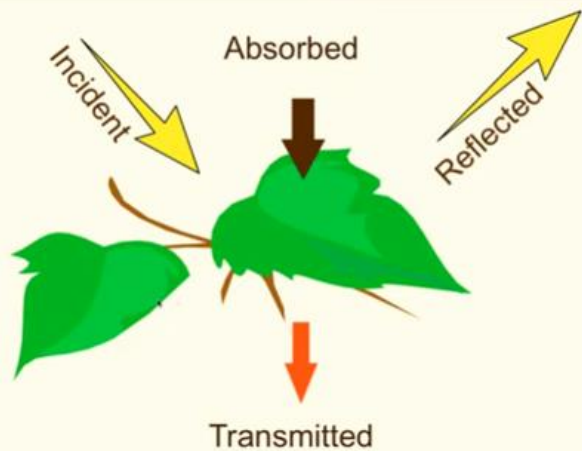
Yellow: 0.578 to 0.592 μm

Orange: 0.592 to 0.620 μm

Red: 0.620 to 0.700 μm

Diagram of LIGHT Interaction

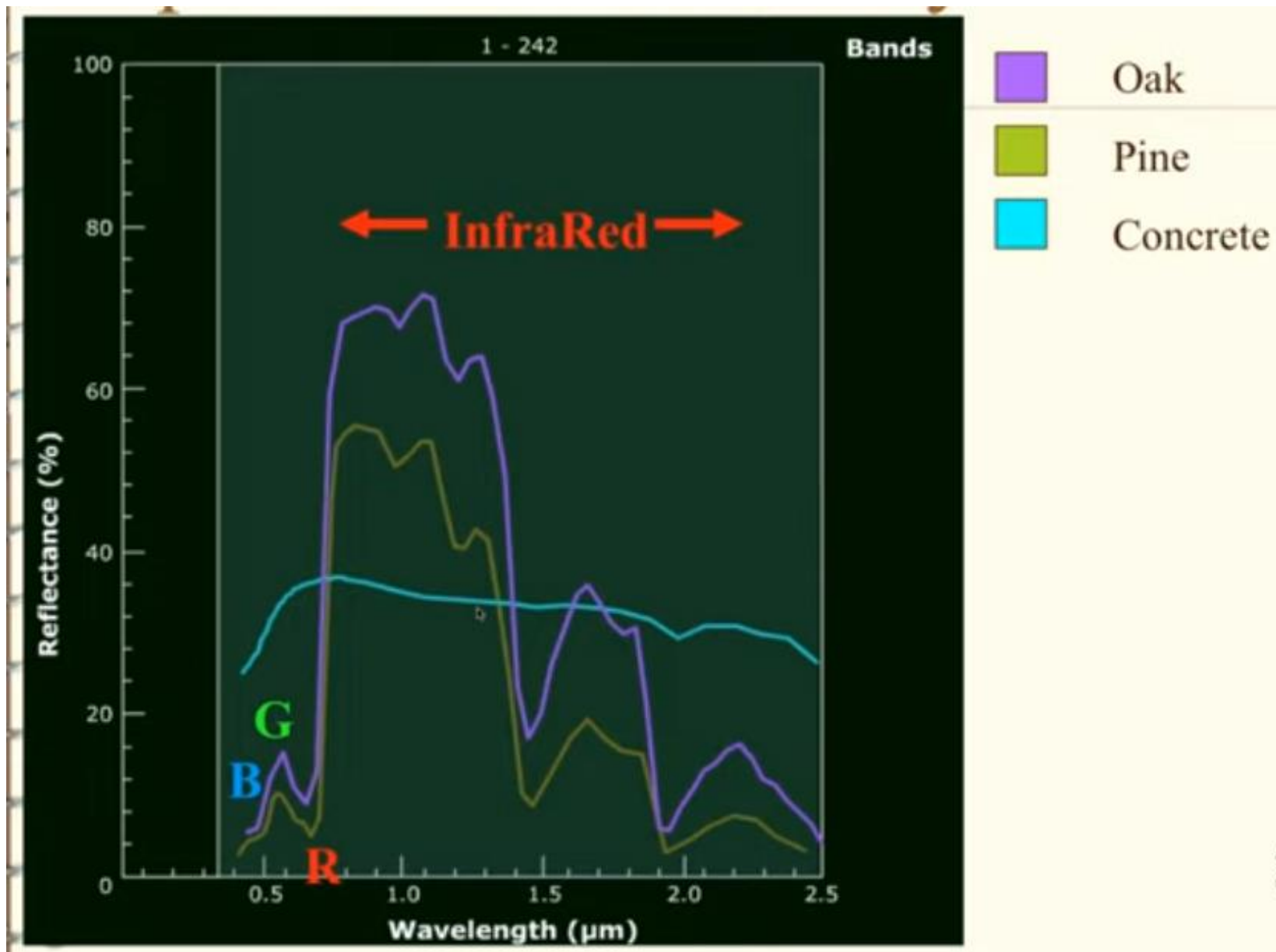
Diagram of Light Interactions



Remote Sensing Basics

9

SPECTRAL PATTERN ANALYSIS



PLATFORMS USED IN REMOTESENSING

- ❑ Ground-based platforms: ground, vehicles, ladders and/or buildings:- up to 50 m
- ❑ Airborne platforms: airplanes, helicopters, high-altitude aircrafts, balloons:- up to 50 km
- ❑ Space borne platforms: rockets, satellites, shuttle:-
from about 100 km to 36000 km
 - Space shuttle: 250 to 300 km
 - Space station: 300 to 400 km
 - Low-level satellites: 700 to 1500 km
 - High-level satellites: about 36000 km



Aircrafts



Ground-based platforms



Satellite

The world's first artificial satellite, Sputnik 1 was launched on 4th October 1957 by Soviet Union, since then, improvements in sensor configuration aboard various earth observing satellites such as Landsat, Nimbus and more recent missions such as RADARSAT and UARS provided global images in different regions of electromagnetic spectrum proved to be useful for various civil, research, and military purposes.

Space probes to other planets also provided an opportunity to carry out studies in extraterrestrial environments. Synthetic aperture radar (SAR) aboard the Magellan spacecraft provided detailed topographic maps of Venus, while sensors aboard SOHO (Solar and Heliospheric Observatory) provided a platform for studies to be performed on the Sun.



The first artificial satellite launched by Soviet Union in 1957



NASA's space shuttle 'Columbia' to monitor Earth resources.

HISTORY OF INDIA'S SPACE PROGRAMME



Dr. Vikram Sarabhai founded the Physical Research Laboratory (PRL) in Ahmedabad on November 11, 1947 that proved to be the first step that India took towards becoming one of the leading space power.

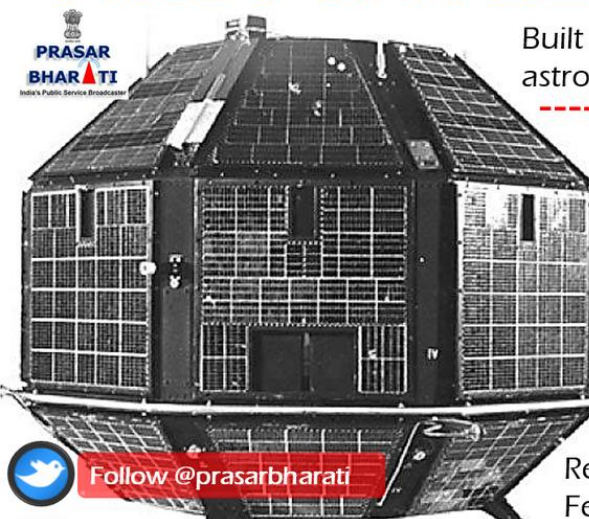
In 1969, The Indian Space Research Organisation (ISRO) was established that superseded the erstwhile Indian National Committee for Space Research (INCOSPAR), which was established in 1962 by Dr. Vikram Sarabhai along with the efforts of independent India's first Prime Minister Jawaharlal Nehru

India's first major success was achieved on April 19, 1975 with the launch of its indigenous satellite named as Aryabhata into space.

It was launched by the Soviet Union from Kapustin Yar using a Cosmos-3M launch vehicle

India's first satellite, named after an ancient Indian mathematician Aryabhata

FIRST SATELLITE OF INDIA



Built by ISRO to conduct experiments in X-ray astronomy, aeronomics, and solar physics

360 Kg spacecraft was a 26-sided polyhedron 1.4 metres (4.6 ft) in diameter

Launched by Soviet Intercosmos rocket on 19 April 1975 from Kapustin Yar.

A power failure halted experiments after 4 days in orbit & signals from the spacecraft were lost after 5 days of operation.

Re-entered the Earth's atmosphere on 11 Feb 1992.

96.3 minute orbit had an apogee of 619 km and a perigee of 563 km, at an inclination of 50.7 degrees

PRASAR BHARATI India's Public Service Broadcaster

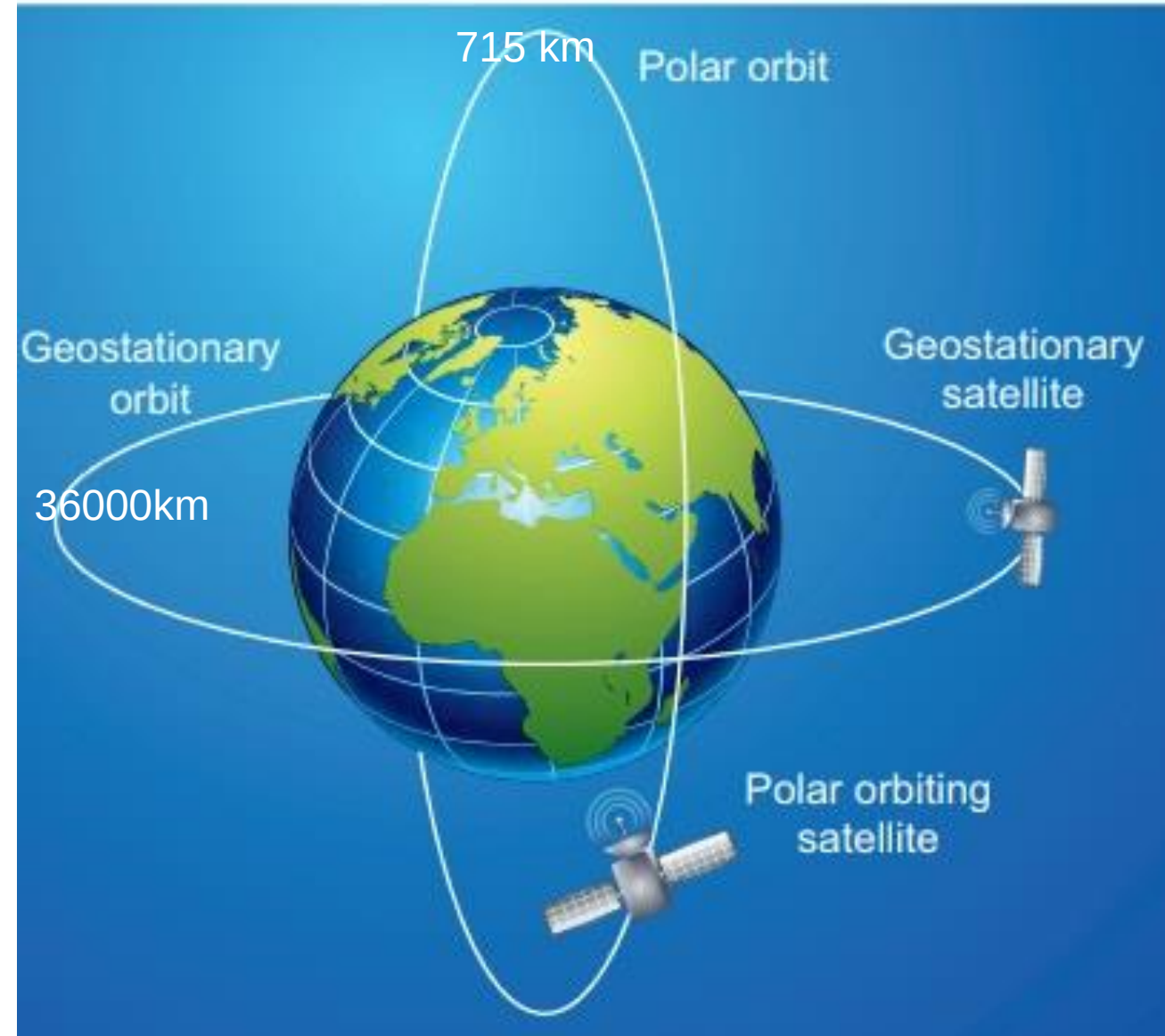
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SATELLITE ROTATING ORBITS

Different types of orbits are required to achieve continuous monitoring (meteorology), global mapping (land cover mapping), or selective imaging (urban areas). The following orbit types are more common for remote sensing

POLAR ORBIT:

Polar Orbit has inclination angle between 80 Degree and 100 Degree that confines the satellite motion in westward direction. Polar orbit enables observation of the whole globe including the poles. Satellites with polar orbit are launched at 600 km to 1000 km altitude.



GEOSTATIONARY ORBIT:

It has 0 (zero degrees) inclination angle i.e. satellite is placed above the equator at an altitude of 36,000 km. The orbital period is kept equal to the rotational period of the earth that results in fixed position of satellite relative to the earth (i.e. satellites observe and collect information continuously over specific areas and always view the same portion of the earth). Due to their high altitude, geostationary weather satellite monitors weather and cloud patterns covering an entire hemisphere of the earth. Geostationary orbits are commonly used for meteorological and telecommunication satellites

SUN-SYNCHRONOUS ORBIT:

Also referred to as near polar orbit, having inclination angle between 90 & 99 degrees which is relative to a line running between the North and South poles thereby enabling the satellite to always pass overhead at the same time. The platforms are designed to adopt an orbit in north-south direction, which, in conjunction with the Earth's rotation (west-east), allows

them to cover most of the Earth's surface over a certain period of time. Most sun synchronous orbits cross the equator at mid morning at around 10:30 hour local sun time. At that moment the sun angle is low and the resultant shadows reveal terrain relief. Examples of sun synchronous satellites are Landsat, IRS and SPOT.

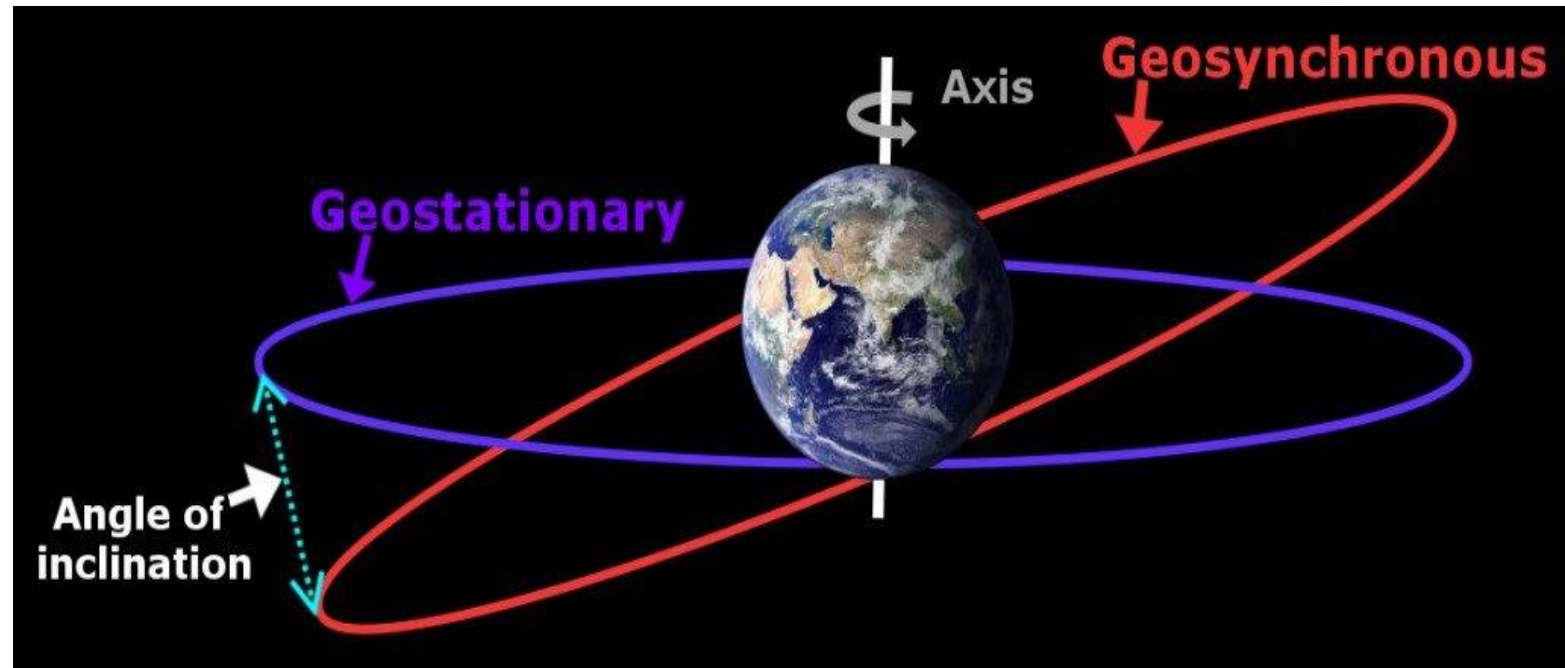
GEOSTATIONARY & GEOSYNCHRONOUS SATELLITE

Definitions of both geostationary and geosynchronous orbits outlined above, it's quite clear that there is very little difference between the two. A satellite in geosynchronous orbit has the same orbital period, i.e., one sidereal day, as that of a satellite in a geostationary orbit.

The only difference between the two is that **while a geosynchronous satellite may or may not be following an inclined orbit (with respect to the equatorial plane), a geostationary satellite has to follow a non-inclined orbit.** In other words, a geostationary satellite remains exactly above the Earth's equator at all times.

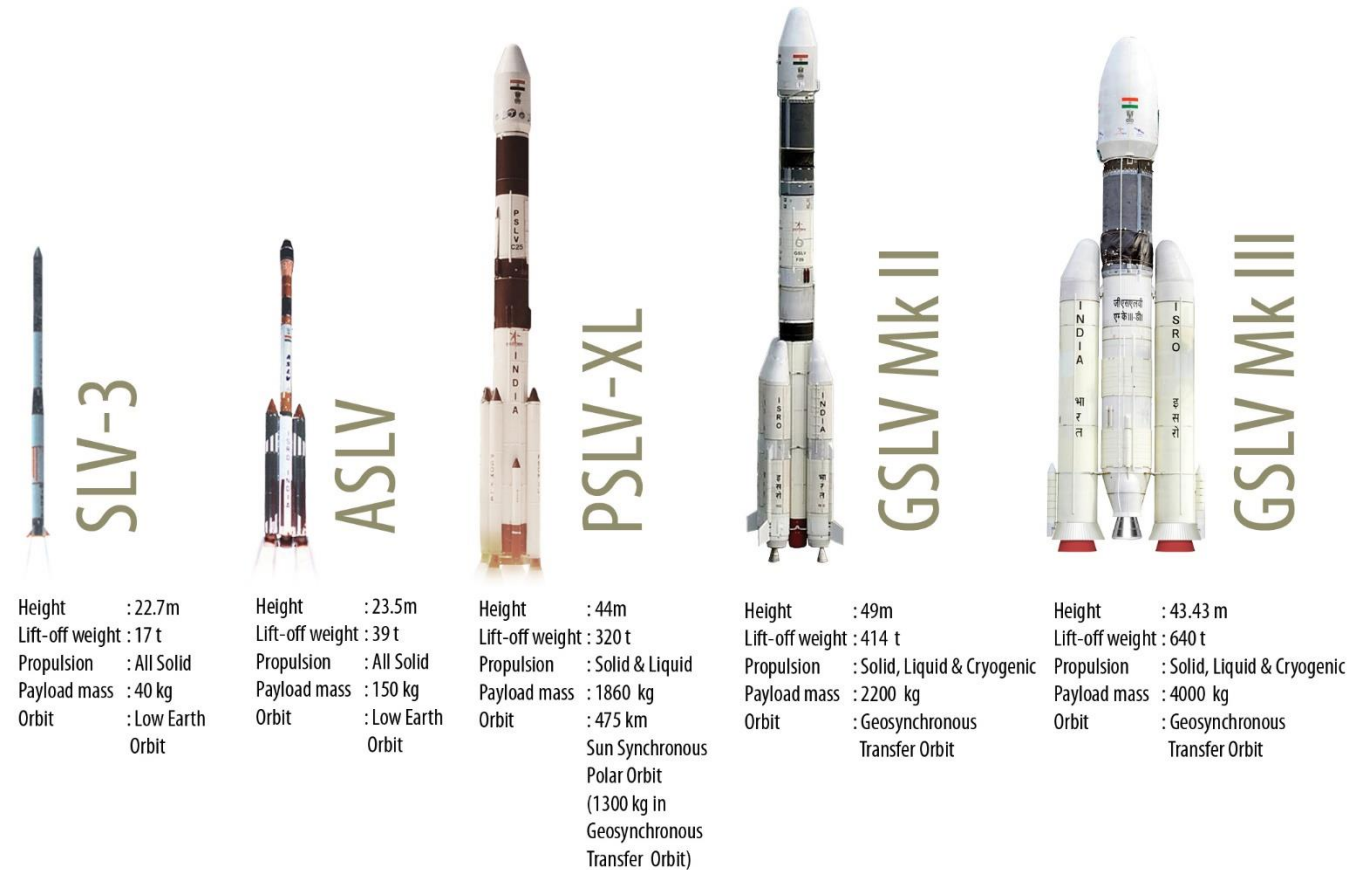
Example :

Russian Raduga 29, Indian Astra 1C, Malaysian MEASAT 2 Geosynchronous satellites are typically used for various purposes, such as communicating back and forth with spacecraft (like the Hubble Space Telescope and space shuttles), voice communication, Internet, broadcasting cable TV and radio signals, as well as weather forecasting.



TYPES OF SATELLITE LAUNCH VEHICLES

- ❑ **SLV** The Satellite Launch Vehicle or SLV was a small-lift launch vehicle project started in the early 1970s by the Indian Space Research Organisation to develop the technology needed to launch satellites. SLV was intended to reach a height of 400 kilometres and carry a payload of 40 kg.
- ❑ **ASLV** Augmented Satellite Launch Vehicle Augmented Satellite Launch Vehicle (**ASLV**) Programme was designed to augment the payload capacity to 150 kg, thrice that of SLV-3, for Low Earth Orbits
- ❑ **GSLV** (Geosynchronous Satellite Launch Vehicle) are the satellite-launch vehicles (rockets) developed by ISRO. This fourth generation launch vehicle is a three stage vehicle with four liquid strap-ons.
- ❑ **PSLV** Polar Satellite Launch Vehicle (PSLV) is the third generation launch vehicle of India. **PSLV** is a four-staged launch vehicle with first and third stage using solid rocket motors and second and fourth stages using liquid rocket engines.



SENSORS

Sensors are the devices used to record the electromagnetic radiations emitted or reflected from the target features and acquire images used in variety of remote sensing applications. In remote sensing, sensors are capable of acquiring information about the target feature to which human eye is insensitive to recognise specially the radiations in other parts of the electromagnetic spectrum than in the visible portion.

Sensors are broadly classified into three categories:

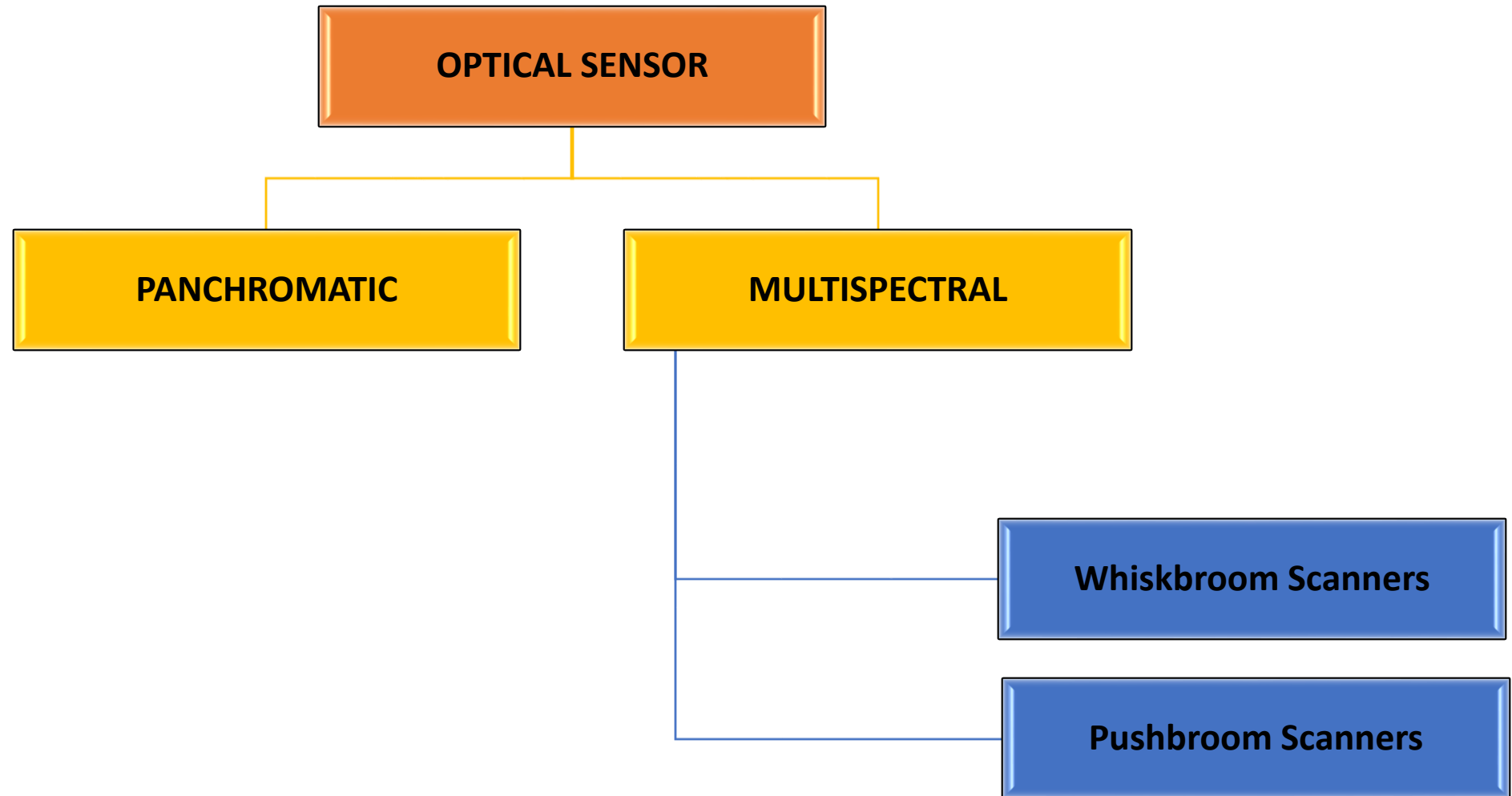
- 1. OPTICAL SENSOR,**
- 2. MICROWAVE SENSOR &**
- 3. THERMAL SENSOR**

OPTICAL SENSOR

- ❑ Optical sensors utilize the energy from the sun which is the source of illumination by recording the energy reflected or emitted from the target feature.
- ❑ Optical sensors record the reflected or emitted energy in the visible, near infrared & short-wave infrared regions of the electromagnetic spectrum.
- ❑ The amount of energy recorded by the sensor depends upon the spectral reflectance characteristics of target features, since, different materials reflect and absorb differently at different wavelengths.
- ❑ For instance, optical sensors are sensitive mainly to the electromagnetic radiations lying in the range of $0.4\ \mu\text{m}$ to $0.76\ \mu\text{m}$ as well as $0.76\ \mu\text{m}$ to $0.9\ \mu\text{m}$ (near and mid infrared band), which are the wavelengths insensitive to the human eye.
- ❑ Images acquired from optical sensors find usefulness in a variety of applications such as post earth quake damage assessment, landslide damage assessment, oil spills, vegetation monitoring, flood assessment and relief measures, land use and land cover classification, temporal change detection analysis and many more.

OPTICAL SENSOR

Optical remote sensing systems are classified into the following types, depending on the number of spectral bands used in the imaging process.



OPTICAL SENSOR

- ❑ Optical sensors utilize the energy from the sun which is the source of illumination by recording the energy reflected or emitted from the target feature.
- ❑ Optical sensors record the reflected or emitted energy in the visible, near infrared and short-wave infrared regions of the electromagnetic spectrum.
- ❑ The amount of energy recorded by the sensor depends upon the spectral reflectance characteristics of target features, since, different materials reflect and absorb differently at different wavelengths. For instance, optical sensors are sensitive mainly to the electromagnetic radiations lying in the range of 0.4 μm to 0.76 μm (visible band) as well as 0.76 μm to 0.9 μm (near and mid infrared band), which are the wavelengths insensitive to the human eye.
- ❑ Images acquired from optical sensors find usefulness in a variety of applications such as post earthquake damage assessment, landslide damage assessment, oil spills, vegetation monitoring, flood assessment and relief measures, land use and land cover classification, temporal change detection analysis and many more.
- ❑ Optical remote sensing systems are classified into the following types, depending on the number of spectral bands used in the imaging process.

OPTICAL SENSOR

Panchromatic imaging system (PAN): The imaging sensor is a single channel detector sensitive to wide range of wavelengths of light typically covering the entire or large portion of visible band of the spectrum thus resulting in grey scaled image (i.e. images containing different shades of black and white). The PAN sensor records or measures the reflectance in terms of apparent brightness of the targets. The spectral information or "colour" of the targets in the resulting image is completely wiped off

Examples of panchromatic imaging systems are:

- ☐ IKONOS PAN
- ☐ SPOT PAN
- ☐ IRS PAN
- ☐ Quick bird PAN



Panchromatic image extracted from SPOT PAN sensor with ground resolution 10 m.
(Source:http://www.crisp.nus.edu.sg/~research/tutorial/opt_int.htm).

MULTISPECTRAL IMAGING SYSTEM

The Multispectral Scanner (MSS) is one of the important Earth observing sensors introduced in the Landsat series of satellites that uses an oscillating mirror to continuously scan the earth surface perpendicular to the spacecraft velocity.

Every mirror sweep scans six lines simultaneously in each of the four spectral bands and images of the objects are recorded across the field of view. The resulting image is a multilayer image which contains both the brightness and spectral (colour) information of the targets being recorded. Examples of multispectral systems are:

- ❑ LANDSAT MSS
- ❑ IRS LISS
- ❑ SPOT XS
- ❑ IKONOS MS

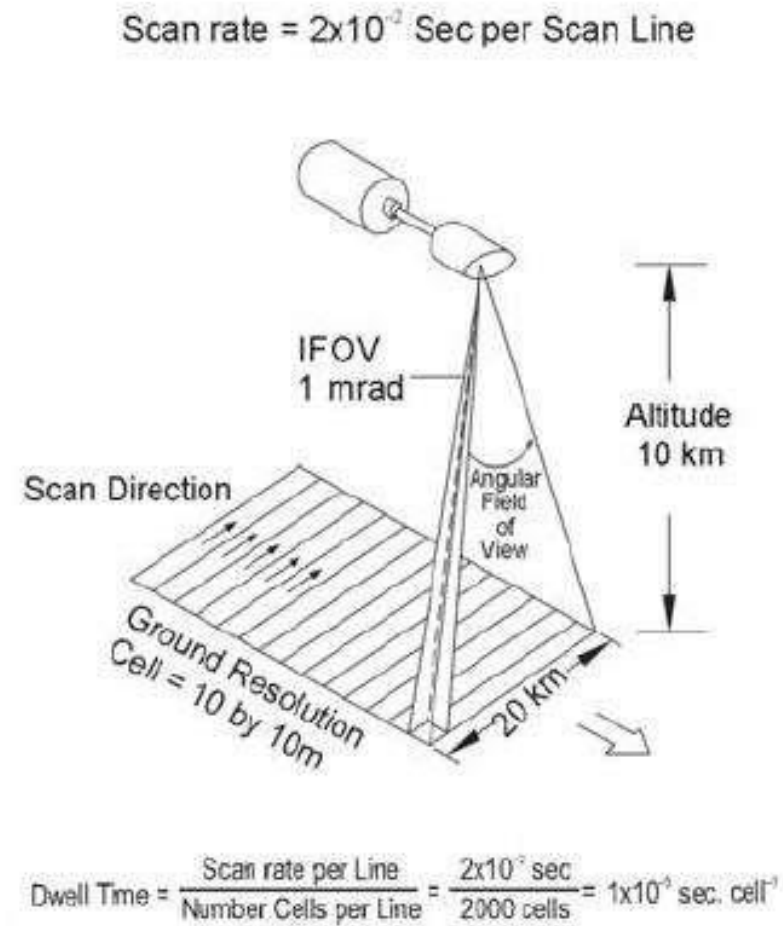
WHISKBROOM SCANNERS:

The whiskbroom scanner is an optical mechanical device which is also known as across track scanner.

These scanners use a rotating mirror and a single detector which scans the scene along a long and narrow band.

The orientation of the mirror is such that on completing one rotation, the detector scans across the field of view between 90° and 120° to obtain images in narrow spectral bands ranging from visible to middle infrared regions of the spectrum.

The angle extended by the mirror in one complete scan known as the Total Field of View (TFOV) of the scanner. Whereas the solid angle extended from a detector to the area on the ground it measures at any instant is termed as Instantaneous Field of View (IFOV)



Whiskbroom scanner

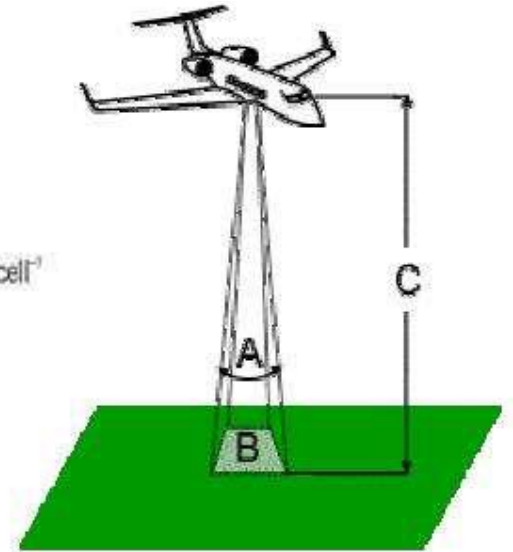


Figure depicts the scanning mechanism of whiskbroom scanners.

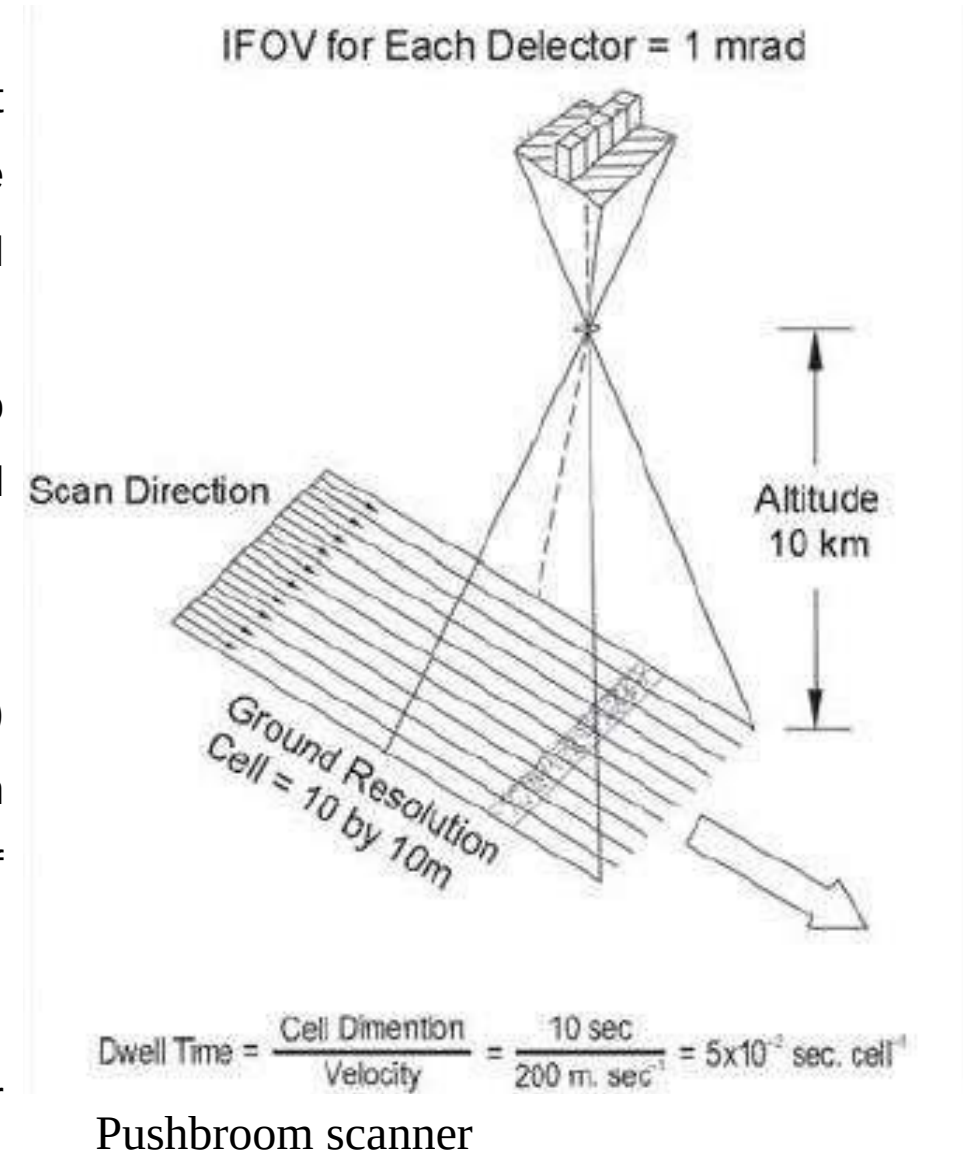
PUSHBROOM SCANNERS:

Also termed as along track scanners and does not have a mirror looking off at varying angles. Instead, there is a line of small sensitive detectors stacked side by side each having tiny dimension on its plate surface; these may be several thousand in number and each detector is a charge coupled device (CCD).

In other words, these scanners consist of a number of detectors equivalent to the swath of the sensor divided by the size of the spatial resolution or pixel size.

For example, the swath of High Resolution Visible Radiometer – 1 (HRVR – 1) of the French remote sensing satellite SPOT is 60 km and the spatial resolution is 20 metres. If we divide 60 km x 1000 metres/20 metres, we get a number of 3000 detectors that are deployed in SPOT HRV – 1 sensor.

The detectors are placed linearly in an arrayed fashion and each detector collects the energy reflected by the ground cell (pixel).



HYPER SPECTRAL IMAGING SYSTEM

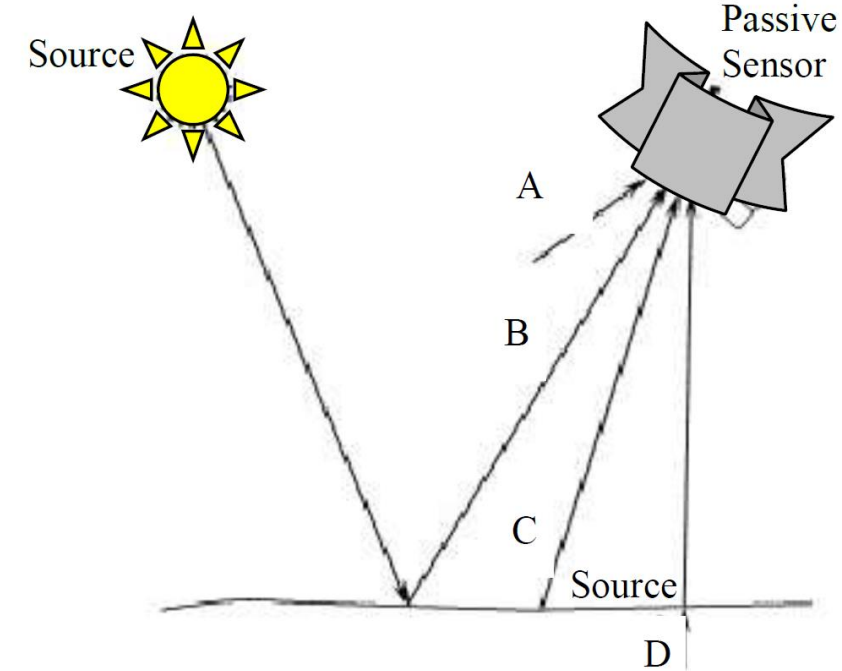
- ❑ Hyperspectral sensors are also referred to as imaging spectrometers that acquire images in a very narrow & large numbers of contiguous spectral bands (i.e. typically collect 200 or more bands of data) throughout the visible, near-IR, mid-IR, and thermal-IR regions of the electromagnetic spectrum

MICROWAVE SENSORS

- ❑ The microwave region of electromagnetic spectrum extends from wavelength approximately 1 cm to 1 m.
- ❑ The longer microwave wavelengths have special characteristics of penetrating the cloud cover, heavy precipitation, haze or dust and are not susceptible to atmospheric scattering which otherwise shorter optical wavelengths are prone to be scattered. These property allows microwave remote sensing in almost all weather and environmental conditions and at any duration of time.
- ❑ In addition, microwave remote sensing also provides useful information on sea wind and wave direction, which are derived from frequency characteristics, Doppler effect, polarization, back scattering etc. that may not be possible by visible and infrared sensors.
- ❑ The sensors operating in the microwave region are broadly classified as **active and passive sensors**.

PASSIVE SENSOR

- ❑ Passive sensors records natural radiation at a particular frequency or range of frequency.
- ❑ The passive microwave sensor records the intensity of the microwave radiation ranging between a frequency range of 5 to 100 GHz emanating from the surface of earth within the antenna's field of view.
- ❑ The passive sensors that measure the emitted energy are the microwave radiometers.
- ❑ The signal recorded through the antenna is represented as an equivalent temperature, in terms of a black body source which would generate equal amount of energy within the bandwidth of the system.
- ❑ Therefore, the microwave energy recorded by a passive sensor is quite low as compared to optical wavelengths since the wavelengths are so long and therefore the field of view of the antenna is kept large to record sufficient energy. Most passive microwave sensors are therefore characterized by low spatial resolution.
- ❑ The important applications of passive microwave remote sensing include areas such as hydrology, oceanography and meteorology.



Microwave energy recorded by a passive sensor

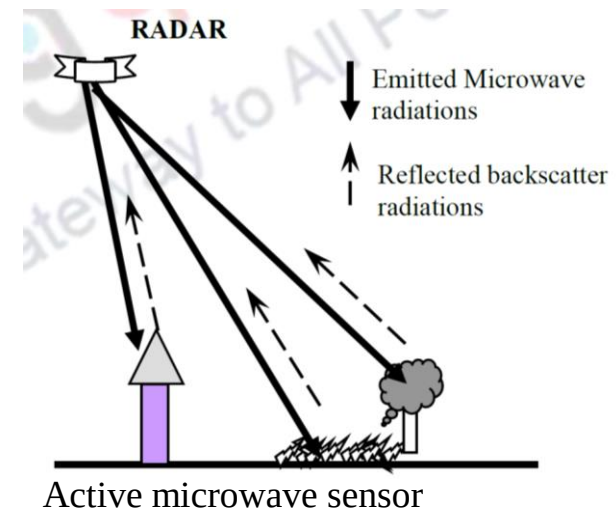
ACTIVE REMOTE SENSOR

emits their own electromagnetic energy in the microwave region toward the surface feature which after interaction with the target produces a backscatter of energy, which is finally recorded by the active sensor's receiver. Active microwave sensors are further categorized into imaging and non-imaging category.

The most common form of imaging active microwave sensors is RADAR which is an acronym for Radio Detection and Ranging. Altimeters and scatterometers come under non-imaging microwave sensors category.

The most widely used active remote sensing system is the RADAR, which transmits long wavelength microwave radiations from 3 cm to 25 cm and records the reflected energy from the target feature in the form of backscatter; LIDAR, an acronym for Light Detection And Ranging is based on the emission and transmission of relatively short wavelength laser waves through the atmosphere and then recording the backscatter radiation;

SONAR, (Sound Navigation And Ranging) is based on the emission and transmission of sound waves through water medium and then recording the backscatter radiation from bottom of the water medium



THERMAL SENSORS

All objects having temperature above absolute zero (0 Kelvin) starts emitting electromagnetic energy in wavelength range between 3 μm to 100 μm . All objects selectively absorb short wavelength solar energy and radiate thermal infrared energy.

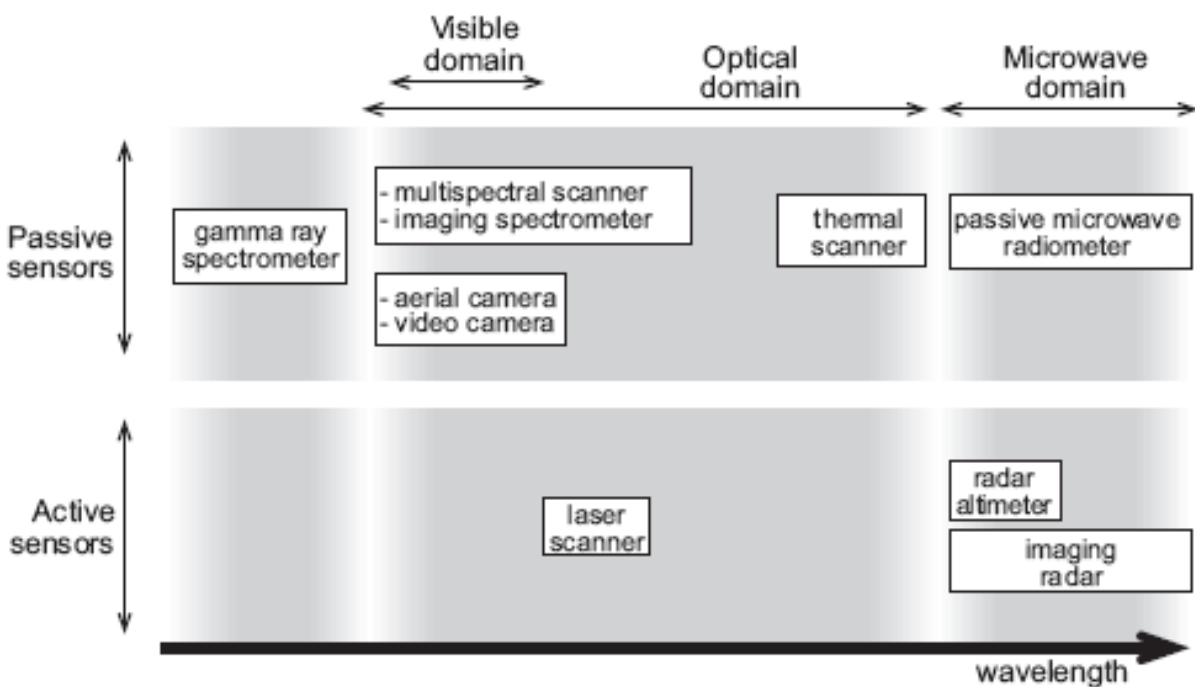
In a thermal image the tone of an object is a function of its surface temperature and its emissivity i.e. all objects emit infrared radiation and the amount of emitted radiation is a function of surface temperature.

Hot objects appear in lighter tone and cooler objects appear darker in an infrared image. In other words, the energy recorded by the radiometer is proportional to the product of the absolute physical temperature (T) and emissivity (ϵ), where ' T ' is referred to as brightness temperature.

As all natural surface features emits radiations so as to keep thermal equilibrium which are measured by radiometer and is represented in terms of a black body.

Applications of thermal infrared remote sensing can be broadly classified into two categories; one in which surface temperature is governed by man made sources of heat and other which is governed by solar radiation. In the former case, the technique has been used from airborne platforms for determining heat losses from buildings and other engineering structures.

In the latter case, thermal infrared remote sensing has been used for identifying crop types, surface soil moisture, monitoring forest fires, military operations as well as identification of crop species for detecting crop diseases.



Overview of sensors

(Source: Kerle, Jamssen and Huurneman; Principles of Remote Sensing)

Land Observing Sensors and their Features

Weather, Global Coverage Satellites

Sensor Name	Pixel Resolution	Swath Width, km	No. Spectral Bands	Spectral Coverage	Temporal Repeat, days
AVHRR	1.1km	2700	5	VNIR, TIR	4*day
SPOT Vegetation	1.15km	2250	4	VNIR, SWIR	26
MODIS	0.25,0.5,1km	2330	36	VNIR, SWIR, TIR	2* day

Regional Satellites

Sensor	m	km	bands	Spectral	Repeat
ASTER	15, 30, 90	60	16	VNIR, SWIR, TIR	16
Landsat TM	30, 120	185	7	VNIR, SWIR, TIR	16
Landsat ETM+	30, 60, 15	185	8	Pan + TM	16
SPOT HRV	10, 20	60	4	Pan, VNIR	26
SPOT HRVIR	10, 20	60	5	SWIR + HRV	26

Local Coverage Satellites

Sensor	m	km	bands	Spectral	Repeat
Quickbird	0.61 Pan, 2.44	16.5	5	Pan, VNIR	2 to 11
IKONOS	1.0 Pan, 4	11.3	5	Pan, VNIR	3

AIRBORNE Instruments

Sensor	m	km	bands	Spectral	Repeat
AVIRIS, Hymap	4, 20	2 km, 10 km	168 - 224	VNIR, SWIR	on demand
CASI-2	5-Jan	1 km - 2.5 km	48-288	VNIR	on demand
ADAR-5500	0.5 - 3	1 km - 2.5 km	4	VNIR	on demand

THANK YOU !